SQL JOIN

fxn

① SELECT employees.employeeID,
employees.FirstName,
employees.LastName,
employees.Salary,
Projects.employeeID,
Projects.Budget
Projects.Status

FROM employees AS INNERJOIN Projects AS B
ON employees.EmployeeID = Projects.employeeID;

EmployeeID	First Name	Last Name	Department	Salary	ProjectName	Budget	Status	Project ID
1	John	Doe	IT	70 000	AI Development	100000	Completed	101
1	John	Doe	IT	70 000				103
2	Alice	Smith	HR	60 000	Employee Training	50 000	on-going	104
3	Bob	Johnson	Finance	75 000	Financial Analysis	90 000	ongoing	104
4	David	Brown	IT	72 000				
5	Emma	Wilson	Sales	65 000	Market Expansion	65 000	Completed	105
6	Michael	Clarke	Finance	80 000	Risk management	80 000	Pending	106

② fxn

```
SELECT employees.employeeID
       employees.FirstName
       employees.LastName,
       employees.Salary,
       employees.Department,
       Projects.ProjectsID,
       Projects.ProjectsName,
       Projects.Budget,
       Projects.Status
FROM Employees AS A LEFT JOIN PROJECTS AS B
ON Employees.EmployeeID = Projects.employeeID;
```

EmployeeID	First Name	Last Name	Department	Salary	ProjectID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
1	John	Doe	IT	70 000	103	Cyber Security Assessment	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee training	50 000	on-going
4	David	Brown	IT	72 000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65 000	105	market expansion	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk management	80 000	Pending
7	Bob	Johnson	Finance	75 000	104	Financial analysis	90 000	on-going

(3) fyn

```
SELECT employees.employeeID,
       employees.FirstName,
       employees.LastName,
       employees.department,
       employees.Salary,
       Projects.Project ID,
       Projects.Project Name,
       Projects.Budget,
       Projects.status
FROM   Projects AS A RIGHTJOIN Employees AS B
ON     Projects.Employee ID = Employees.Employee ID;
```

ProjectsID	Projects Name	Budget	status	Employee ID	First Name	Last Name	Department	Salary
101	AI Development	100 000	Completed	1	John	Doe	IT	70 000
102	Employee training	50 000	ongoing	2	Alice	Smith	HR	60 000
103	Cyber Security audit	75 000	Pending	1	John	Doe	IT	70 000
104	Financial analysis	90 000	ongoing	3	Bob	Johnson	Finance	75 000
105	Market expansion	65 000	Completed	5	Emma	Wilson	Sales	65 000
106	Risk management	80 000	Pending	6	Michael	Clark	Finance	80 000

(4) fxn


```
SELECT employees.employee ID,
       employees.FirstName,
       employees.LastName,
       employees.Department,
       employees.Salary,
       Projects.Project ID,
       Projects.Project Name,
       Projects.Budget,
       Projects.Status
FROM employees AS A FULL OUTER JOIN Projects AS B
ON employees.employeeID = Projects.employee ID;
```

Employee ID	First Name	Last Name	Department	Salary	Projects ID	Project Name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
2	Alice	Smith	HR	60 000	102	Employee training	50 000	ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Expansion	90 000	ongoing
4	David	Brown	IT	72 000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65 000	105	market expansion	65 000	completed
6	Michae	Clark	Finance	80 000	106	Risk management	80 000	Pending

(5) fxn P.5

```
SELECT City AS 'Location'
FROM Employees
UNION
SELECT status
FROM Projects;
```

Location	Status
New York	Pending
Los Angeles	on-going
Toronto	on-going
London	NULL
Sydney	Completed

Filtering


```
SELECT Employee ID,
       First Name,
       Last Name,
       Department,
       Salary
FROM   Employees
WHERE  Salary > 70 000;
```

Employee ID	First Name	Last Name	Department	Salary
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

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```
SELECT Employee ID,
       First Name,
       Last Name,
       Department,
       Salary
FROM Employees
WHERE Department = 'IT' OR 'Finance'
```

Employee ID	First Name	Last Name	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

~~P70~~

```
SELECT  Project ID,
        Project Name,
        Budget,
        Status
FROM    Projects
WHERE   Status = 'Pending' AND 'on-going';
```

Project ID	Project Name	Budget	Status
102	Employee training	50 000	on-going
103	Cyber Security audit	75 000	Pending
104	Financial analysis	90 000	on-going
106	Risk management	80 000	Pending

10)


```
SELECT Project ID,
       Project Name,
       Budget,
       Status
FROM Projects
WHERE Budget 770 000 AND Status = 'ongoing' AND 'Pending';
```

Project ID	Project Name	Budget	Status
103	Cyber security audit	75000	Pending
106	Risk Management	80 000	Pending
104	Financial Analysis	90 000	Ongoing

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```
SELECT Employee ID,
       First Name,
       Last Name,
       Department,
       Salary
FROM Employees
ORDER BY salary DESC
LIMIT 3;
```

Employee ID	First Name	Last Name	Department	Salary
6	Michael	Clark	Finance	80 000
3	Bob	Johnson	Finance	75 000
4	~~Johnson~~ David	Brown	IT	72 000